VIT-AP UNIVERSITY, ANDHRA PRADESH

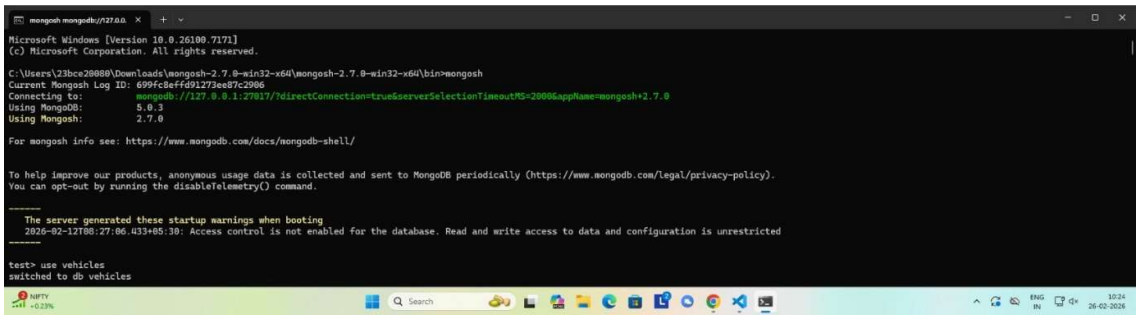
Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech **Date: 26-02-2026**
Faculty Name: Prof. S.Gopikrishnan **School: SCOPE**
Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**

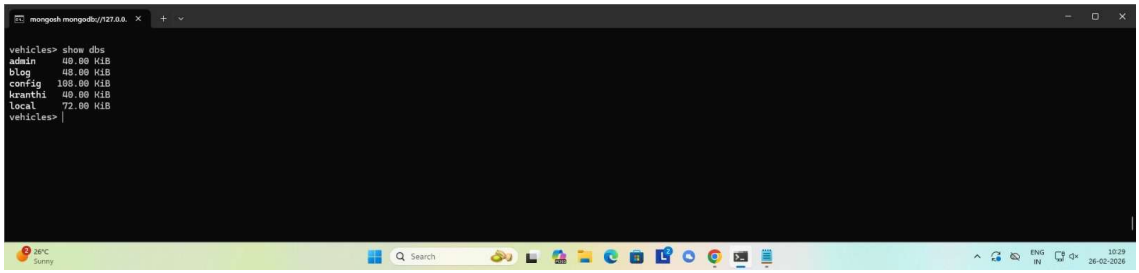
---

1. Use MongoDB to implement the following DB operations

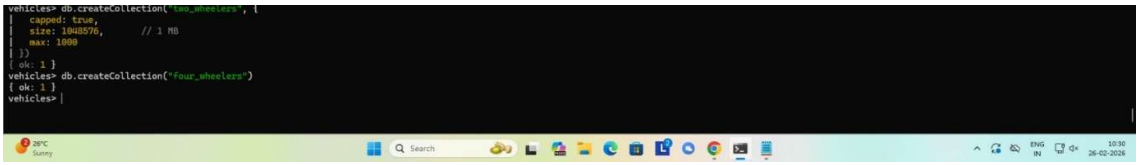
   1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".



   2. Write a MongoDB query to display all the databases.



   3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.



   4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

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Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**

```
vehicles> db.two_wheelers.insertMany([
  {
    bike_name: "Honda Activa 6G",
    model: "gearless",
    category: "125cc",
    colors_available: ["Pearl Precious White", "Rebel Red Metallic", "Mat Axis Grey Metallic", "Midnight Black Metallic"],
    manufacturer: "Honda",
    performance: 8,
    timestamp: ISODate("2020-06-01T00:00:00Z"),
    price: 78000
  },
  {
    bike_name: "Bajaj Pulsar NS200",
    model: "gear",
    category: "200cc",
    colors_available: ["Pearl Metallic White", "Burnt Red", "Graphite Black", "Satin Silver"],
    manufacturer: "Bajaj",
    performance: 9,
    timestamp: ISODate("2012-07-15T00:00:00Z"),
    price: 152000
  },
  {
    bike_name: "Royal Enfield Classic 350",
    model: "gear",
    category: "350cc",   // bonus category
    colors_available: ["Halcyon Green", "Gunmetal Grey", "Dark Stealth Black"],
    manufacturer: "Royal Enfield",
    performance: 7.5,
    timestamp: ISODate("2021-09-01T00:00:00Z"),
    price: 195000
  },
  {
    bike_name: "TVS Apache RTR 160",
    model: "gear",
    category: "160cc",
    colors_available: ["Racing Red", "Pearl White", "Matt Blue", "Titanium Grey"],
    manufacturer: "TVS",
    performance: 8.5,
```

```
    bike_name: "TVS Apache RTR 160",
    model: "gear",
    category: "160cc",
    colors_available: ["Racing Red", "Pearl White", "Matt Blue", "Titanium Grey"],
    manufacturer: "TVS",
    performance: 8.5,
    timestamp: ISODate("2019-11-20T00:00:00Z"),
    price: 128000
  },
  {
    bike_name: "Hero Splendor Plus",
    model: "gearless",
    category: "100cc",
    colors_available: ["Black Nexus Blue", "Heavy Grey with Green", "Black Heavy Grey with Green"],
    manufacturer: "Hero",
    performance: 6,
    timestamp: ISODate("2022-03-10T00:00:00Z"),
    price: 75000
  }
])

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699fd392fd91273ee87c2911'),
    '1': ObjectId('699fd392fd91273ee87c2912'),
    '2': ObjectId('699fd392fd91273ee87c2913'),
    '3': ObjectId('699fd392fd91273ee87c2914'),
    '4': ObjectId('699fd392fd91273ee87c2915')
  }
}
vehicles>
```

5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

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Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**

```
vehicles> db.two_wheelers.insertMany([
  {
    bike_name: "Honda Activa 6G",
    model: "gearless",
    category: "125cc",
    colors_available: ["Pearl Precious White", "Rebel Red Metallic", "Mat Axis Grey Metallic", "Midnight Black Metallic"],
    manufacturer: "Honda",
    performance: 8,
    timestamp: ISODate("2020-06-01T00:00:00Z"),
    price: 78000
  },
  {
    bike_name: "Bajaj Pulsar NS200",
    model: "gear",
    category: "200cc",
    colors_available: ["Pearl Metallic White", "Burnt Red", "Graphite Black", "Satin Silver"],
    manufacturer: "Bajaj",
    performance: 9,
    timestamp: ISODate("2012-07-15T00:00:00Z"),
    price: 152000
  },
  {
    bike_name: "Royal Enfield Classic 350",
    model: "gear",
    category: "350cc",   // bonus category
    colors_available: ["Halcyon Green", "Gunmetal Grey", "Dark Stealth Black"],
    manufacturer: "Royal Enfield",
    performance: 7.5,
    timestamp: ISODate("2021-09-01T00:00:00Z"),
    price: 195000
  },
  {
    bike_name: "TVS Apache RTR 160",
    model: "gear",
    category: "160cc",
    colors_available: ["Racing Red", "Pearl White", "Matt Blue", "Titanium Grey"],
    manufacturer: "TVS",
    performance: 8.5,
```

```
    vehicle_name: "Eicher Pro 2049",
    model: "commercial",
    category: "mini truck",
    variants: ["14 Feet", "17 Feet", "Diesel"],
    manufacturer: "Eicher",
    performance: 6.5,
    timestamp: ISODate("2018-06-01T00:00:00Z"),
    price: 1650000
  },
  {
    vehicle_name: "Ashok Leyland Dost+",
    model: "commercial",
    category: "mini truck",
    variants: ["LX", "S2", "Diesel"],
    manufacturer: "Ashok Leyland",
    performance: 6,
    timestamp: ISODate("2021-04-01T00:00:00Z"),
    price: 780000
  }
])

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699fd42efd91273ee87c2916'),
    '1': ObjectId('699fd42efd91273ee87c2917'),
    '2': ObjectId('699fd42efd91273ee87c2918'),
    '3': ObjectId('699fd42efd91273ee87c2919'),
    '4': ObjectId('699fd42efd91273ee87c291a')
  }
}
```

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.

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Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**

```
vehicles> db.two_wheelers.find().pretty()
[
  {
    _id: ObjectId('699fd392fd91273ee87c2911'),
    bike_name: 'Honda Activa 6G',
    model: 'gearless',
    category: '125cc',
    colors_available: [
      'Pearl Precious White',
      'Rebel Red Metallic',
      'Nut Axis Grey Metallic',
      'Midnight Black Metallic'
    ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: ISODate('2020-06-01T00:00:00.000Z'),
    price: 78000
  },
  {
    _id: ObjectId('699fd392fd91273ee87c2912'),
    bike_name: 'Bajaj Pulsar NS200',
    model: 'gear',
    category: '200cc',
    colors_available: [
      'Pearl Metallic White',
      'Burnt Red',
      'Graphite Black',
      'Satin Silver'
    ],
    manufacturer: 'Bajaj',
    performance: 9,
```

```
    ],
    manufacturer: 'Bajaj',
    perfromance: 9,
    timestamp: ISODate('2012-07-15T00:00:00.000Z'),
    price: 152000
  },
  {
    _id: ObjectId('699fd392fd91273ee87c2913'),
    bike_name: 'Royal Enfield Classic 350',
    model: 'gear',
    category: '350cc',
    colors_available: [ 'Halcyon Green', 'Gunmetal Grey', 'Dark Stealth Black' ],
    manufacturer: 'Royal Enfield',
    performance: 7.5,
    timestamp: ISODate('2021-09-01T00:00:00.000Z'),
    price: 195000
  },
  {
    _id: ObjectId('699fd392fd91273ee87c2914'),
    bike_name: 'TVS Apache RTR 160',
    model: 'gear',
    category: '160cc',
    colors_available: [ 'Racing Red', 'Pearl White', 'Matt Blue', 'Titanium Grey' ],
    manufacturer: 'TVS',
    performance: 8.5,
    timestamp: ISODate('2019-11-20T00:00:00.000Z'),
    price: 128000
  },
  {
    _id: ObjectId('699fd392fd91273ee87c2915'),
    bike_name: 'Hero Splendor Plus',
    model: 'gearless',
    category: '100cc',
```

```
    model: 'gearless',
    category: '100cc',
    colors_available: [
      'Black Nexus Blue',
      'Heavy Grey with Green',
      'Black Heavy Grey with Green'
    ],
    manufacturer: 'Hero',
    performance: 6,
    timestamp: ISODate('2022-03-10T00:00:00.000Z'),
    price: 75000
  }
]
```

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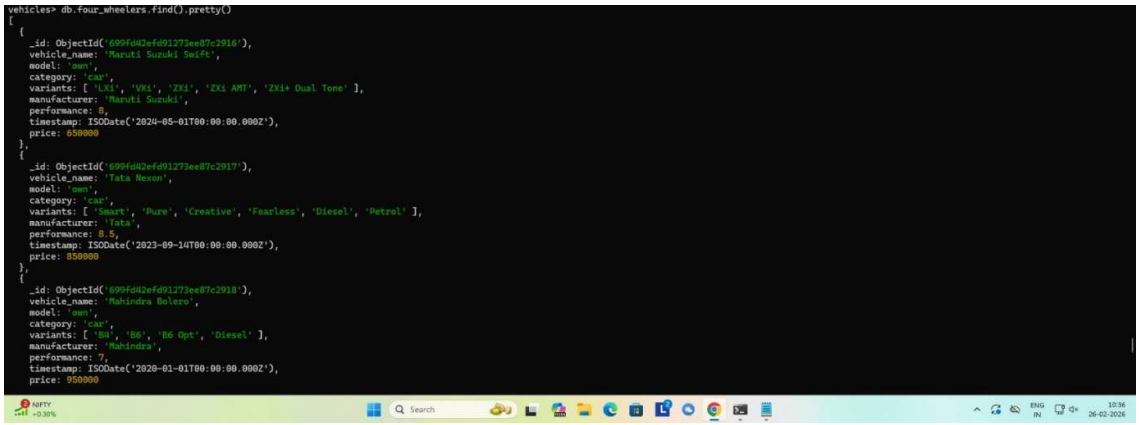
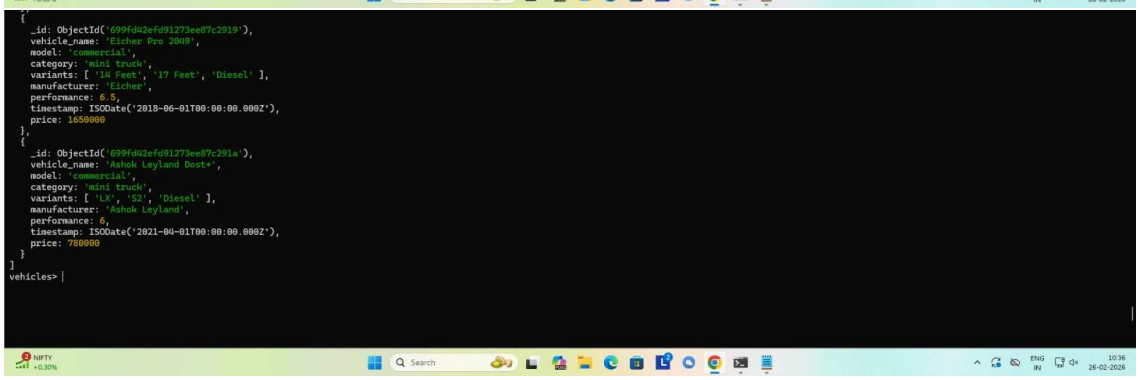
Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**

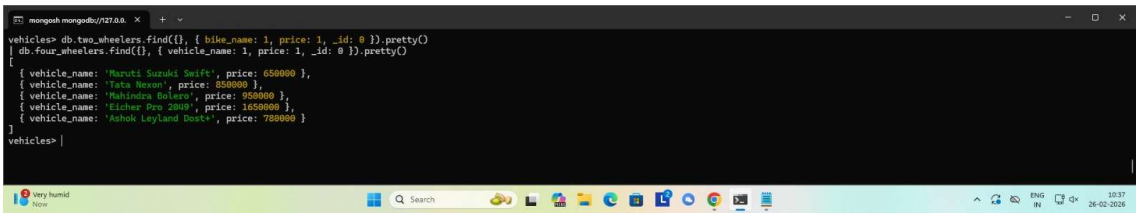


```
vehicles> db.four_wheelers.find().pretty()
[
  {
    _id: ObjectId('699fd42efd91273ee87c2916'),
    vehicle_name: 'Maruti Suzuki Swift',
    model: 'own',
    category: 'car',
    variants: [ 'LXi', 'VXi', 'ZXi', 'ZXi AMT', 'ZXi+ Dual Tone' ],
    manufacturer: 'Maruti Suzuki',
    performance: 8,
    timestamp: ISODate('2024-05-01T00:00:00.000Z'),
    price: 650000
  },
  {
    _id: ObjectId('699fd42efd91273ee87c2917'),
    vehicle_name: 'Tata Nexon',
    model: 'own',
    category: 'car',
    variants: [ 'Smart', 'Pure', 'Creative', 'Fearless', 'Diesel', 'Petrol' ],
    manufacturer: 'Tata',
    performance: 8.5,
    timestamp: ISODate('2023-09-14T00:00:00.000Z'),
    price: 850000
  },
  {
    _id: ObjectId('699fd42efd91273ee87c2918'),
    vehicle_name: 'Mahindra Bolero',
    model: 'own',
    category: 'car',
    variants: [ 'B4', 'B6', 'B6 Opt', 'Diesel' ],
    manufacturer: 'Mahindra',
    performance: 7,
    timestamp: ISODate('2020-01-01T00:00:00.000Z'),
    price: 950000
```



```
  {
    _id: ObjectId('699fd42efd91273ee87c2919'),
    vehicle_name: 'Eicher Pro 2049',
    model: 'commercial',
    category: 'mini truck',
    variants: [ '14 Feet', '17 Feet', 'Diesel' ],
    manufacturer: 'Eicher',
    performance: 6.5,
    timestamp: ISODate('2018-06-01T00:00:00.000Z'),
    price: 1650000
  },
  {
    _id: ObjectId('699fd42efd91273ee87c291a'),
    vehicle_name: 'Ashok Leyland Dost+',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'LX', 'S2', 'Diesel' ],
    manufacturer: 'Ashok Leyland',
    performance: 6,
    timestamp: ISODate('2021-04-01T00:00:00.000Z'),
    price: 780000
  }
]
vehicles>
```

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database



```
vehicles> db.two_wheelers.find({}, { bike_name: 1, price: 1, _id: 0 }).pretty()
| db.four_wheelers.find({}, { vehicle_name: 1, price: 1, _id: 0 }).pretty()
[
  { vehicle_name: 'Maruti Suzuki Swift', price: 650000 },
  { vehicle_name: 'Tata Nexon', price: 850000 },
  { vehicle_name: 'Mahindra Bolero', price: 950000 },
  { vehicle_name: 'Eicher Pro 2049', price: 1650000 },
  { vehicle_name: 'Ashok Leyland Dost+', price: 780000 }
]
vehicles>
```

8. Write a MongoDB query to display two_wheelers from a particular company

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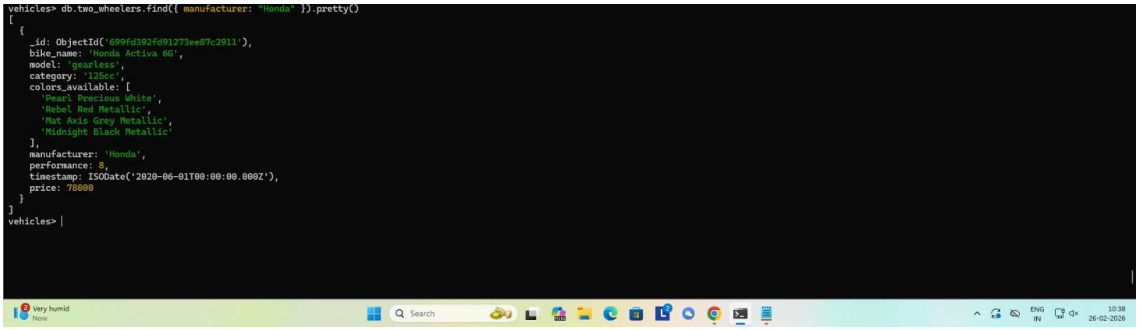
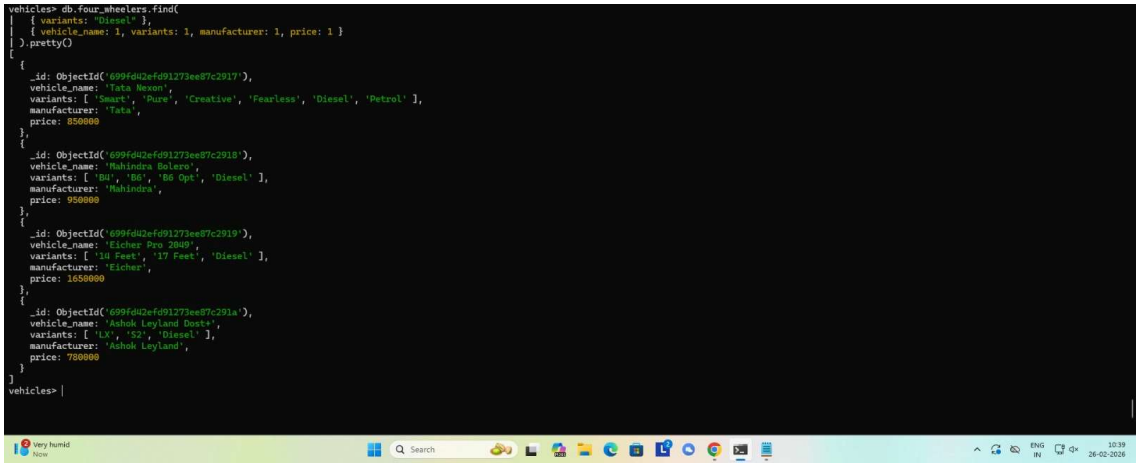


```
vehicles> db.two_wheelers.find({ manufacturer: "Honda" }).pretty()
[
  {
    _id: ObjectId('699fd392fd91273ee87c2911'),
    bike_name: 'Honda Activa 6G',
    model: 'gearless',
    category: '125cc',
    colors_available: [
      'Pearl Precious White',
      'Rebel Red Metallic',
      'Mat Axis Grey Metallic',
      'Midnight Black Metallic'
    ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: ISODate('2020-06-01T00:00:00.000Z'),
    price: 78000
  }
]
vehicles>
```

9. Write a MongoDB query to display four_wheelers available in diesel variants



```
vehicles> db.four_wheelers.find(
|   { variants: "Diesel" },
|   { vehicle_name: 1, variants: 1, manufacturer: 1, price: 1 }
| ).pretty()
[
  {
    _id: ObjectId('699fd42efd91273ee87c2917'),
    vehicle_name: 'Tata Nexon',
    variants: [ 'Smart', 'Pure', 'Creative', 'Fearless', 'Diesel', 'Petrol' ],
    manufacturer: 'Tata',
    price: 850000
  },
  {
    _id: ObjectId('699fd42efd91273ee87c2918'),
    vehicle_name: 'Mahindra Bolero',
    variants: [ 'B4', 'B6', 'B6 Opt', 'Diesel' ],
    manufacturer: 'Mahindra',
    price: 950000
  },
  {
    _id: ObjectId('699fd42efd91273ee87c2919'),
    vehicle_name: 'Eicher Pro 2049',
    variants: [ '14 Feet', '17 Feet', 'Diesel' ],
    manufacturer: 'Eicher',
    price: 1650000
  },
  {
    _id: ObjectId('699fd42efd91273ee87c291a'),
    vehicle_name: 'Ashok Leyland Dost+',
    variants: [ 'LX', 'S2', 'Diesel' ],
    manufacturer: 'Ashok Leyland',
    price: 780000
  }
]
vehicles>
```

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

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Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**

```
vehicles> db.two_wheelers.aggregate([
|   {
|     $project: {
|       name: "$bike_name",
|       category: 1,
|       manufacturer: 1,
|       performance: 1,
|       type: "two_wheeler"
|     }
|   },
|   {
|     $unionWith: {
|       coll: "four_wheelers",
|       pipeline: [
|         {
|           $project: {
|             name: "$vehicle_name",
|             category: 1,
|             manufacturer: 1,
|             performance: 1,
|             type: "four_wheeler"
|           }
|         }
|       ]
|     }
|   },
|   { $match: { performance: { $gt: 5 } } },
|   { $project: { name: 1, category: 1, manufacturer: 1, performance: 1, _id: 0 } }
| ]).pretty()
```

```
[
  {
    category: '125cc',
    manufacturer: 'Honda',
    performance: 8,
    name: 'Honda Activa 6G'
  },
  {
    category: '200cc',
    manufacturer: 'Bajaj',
    performance: 9,
    name: 'Bajaj Pulsar NS200'
  },
  {
    category: '350cc',
    manufacturer: 'Royal Enfield',
    performance: 7.5,
    name: 'Royal Enfield Classic 350'
  },
  {
    category: '160cc',
    manufacturer: 'TVS',
    performance: 8.5,
    name: 'TVS Apache RTR 160'
  },
  {
    category: '100cc',
    manufacturer: 'Hero',
    performance: 6,
    name: 'Hero Splendor Plus'
  },
  {
    category: 'car',
    manufacturer: 'Maruti Suzuki',
    performance: 8,
    name: 'Maruti Suzuki Swift'
```

```
  {
    category: 'car',
    manufacturer: 'Tata',
    performance: 8.5,
    name: 'Tata Nexon'
  },
  {
    category: 'car',
    manufacturer: 'Mahindra',
    performance: 7,
    name: 'Mahindra Bolero'
  },
  {
    category: 'mini truck',
    manufacturer: 'Eicher',
    performance: 6.5,
    name: 'Eicher Pro 2049'
  },
  {
    category: 'mini truck',
    manufacturer: 'Ashok Leyland',
    performance: 6,
    name: 'Ashok Leyland Dost+'
  }
]
vehicles>
```

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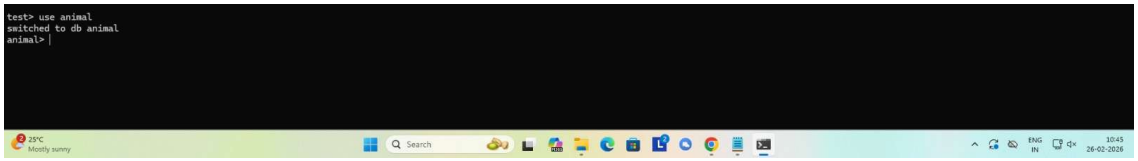
Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**

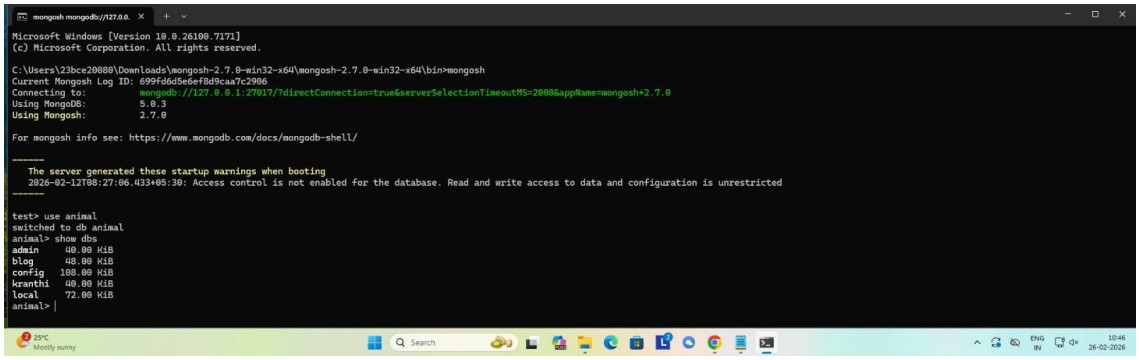
---

2. Use MongoDB to implement the following DB operations for a Zoo

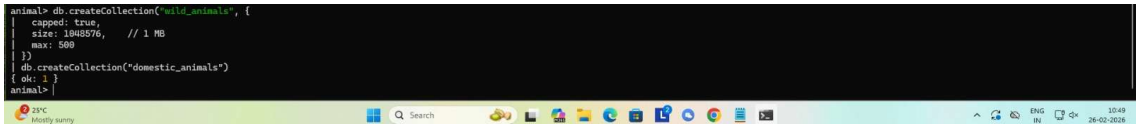
1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.



2. Write a MongoDB query to display all the databases.



3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.



4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

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Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**

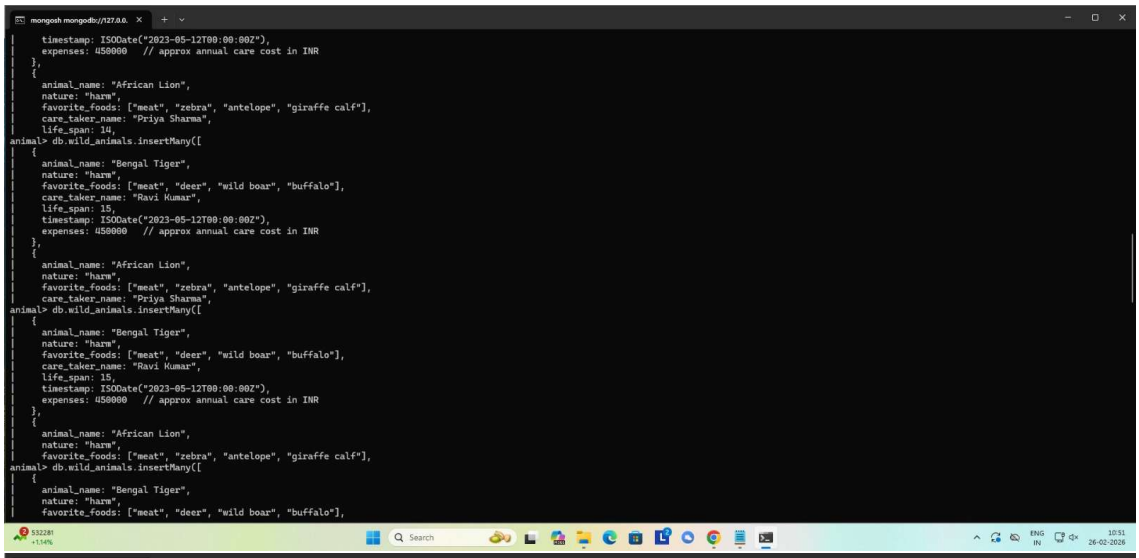
```
mongosh mongodb://127.0.0.
{ ok: 1 }
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
|     expenses: 450000   // approx annual care cost in INR
|   },
|   {
|     animal_name: "African Lion",
|     nature: "harm",
|     favorite_foods: ["meat", "zebra", "antelope", "giraffe calf"],
|     care_taker_name: "Priya Sharma",
|     life_span: 14,
|     timestamp: ISODate("2022-11-03T00:00:00Z"),
|     expenses: 520000
|   },
|   {
|     animal_name: "Indian Elephant",
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
|     expenses: 450000   // approx annual care cost in INR
|   },
|   {
|     animal_name: "African Lion",
|     nature: "harm",
|     favorite_foods: ["meat", "zebra", "antelope", "giraffe calf"],
|     care_taker_name: "Priya Sharma",
|     life_span: 14,
|     timestamp: ISODate("2022-11-03T00:00:00Z"),
|     expenses: 520000
|   },
|   {
animal> db.wild_animals.insertMany([
```

```
mongosh mongodb://127.0.0.
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
|     expenses: 450000   // approx annual care cost in INR
|   },
|   {
|     animal_name: "African Lion",
|     nature: "harm",
|     favorite_foods: ["meat", "zebra", "antelope", "giraffe calf"],
|     care_taker_name: "Priya Sharma",
|     life_span: 14,
|     timestamp: ISODate("2022-11-03T00:00:00Z"),
|     expenses: 520000
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
|     expenses: 450000   // approx annual care cost in INR
|   },
|   {
|     animal_name: "African Lion",
|     nature: "harm",
|     favorite_foods: ["meat", "zebra", "antelope", "giraffe calf"],
|     care_taker_name: "Priya Sharma",
|     life_span: 14,
|     timestamp: ISODate("2022-11-03T00:00:00Z"),
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
```

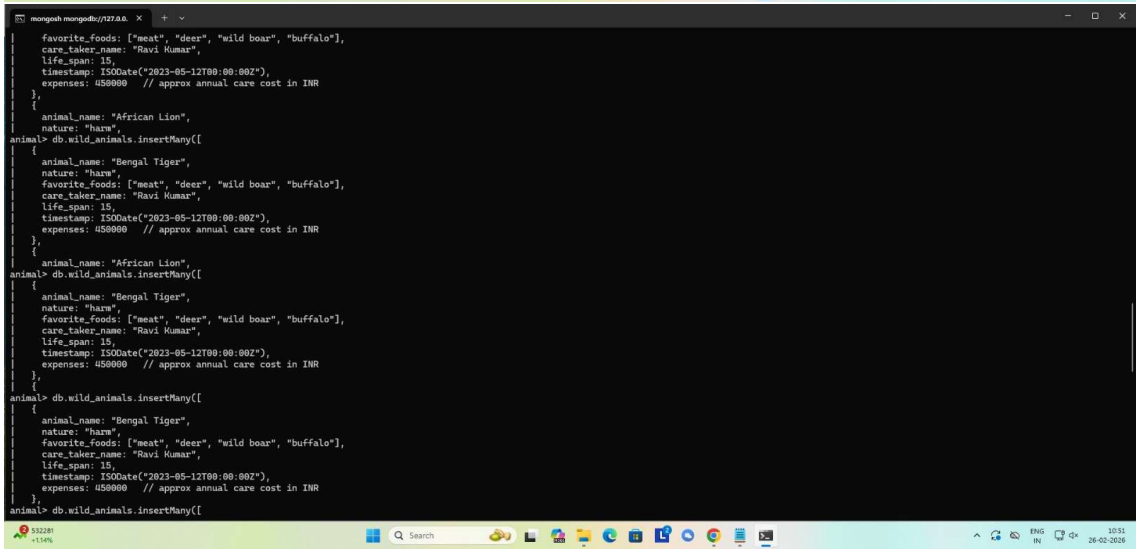
VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech **Date: 26-02-2026**
Faculty Name: Prof. S.Gopikrishnan **School: SCOPE**
Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**



```
mongosh mongodb://127.0.0.
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
|     expenses: 450000   // approx annual care cost in INR
|   },
|   {
|     animal_name: "African Lion",
|     nature: "harm",
|     favorite_foods: ["meat", "zebra", "antelope", "giraffe calf"],
|     care_taker_name: "Priya Sharma",
|     life_span: 14,
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
|     expenses: 450000   // approx annual care cost in INR
|   },
|   {
|     animal_name: "African Lion",
|     nature: "harm",
|     favorite_foods: ["meat", "zebra", "antelope", "giraffe calf"],
|     care_taker_name: "Priya Sharma",
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
|     expenses: 450000   // approx annual care cost in INR
|   },
|   {
|     animal_name: "African Lion",
|     nature: "harm",
|     favorite_foods: ["meat", "zebra", "antelope", "giraffe calf"],
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
```

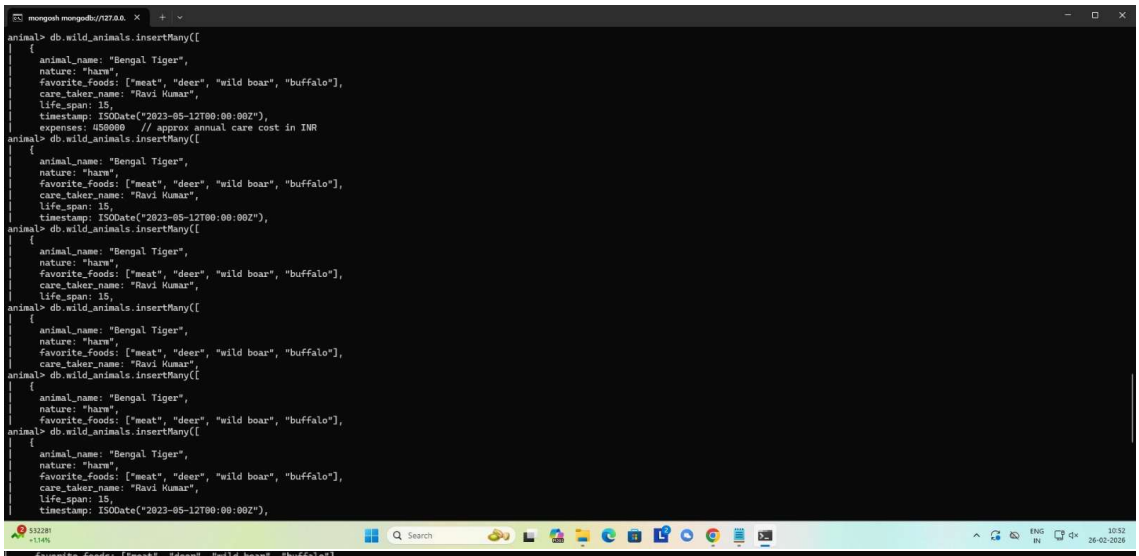


```
mongosh mongodb://127.0.0.
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
|     expenses: 450000   // approx annual care cost in INR
|   },
|   {
|     animal_name: "African Lion",
|     nature: "harm",
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
|     expenses: 450000   // approx annual care cost in INR
|   },
|   {
|     animal_name: "African Lion",
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
|     expenses: 450000   // approx annual care cost in INR
|   },
|   {
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
|     expenses: 450000   // approx annual care cost in INR
|   },
animal> db.wild_animals.insertMany([
```

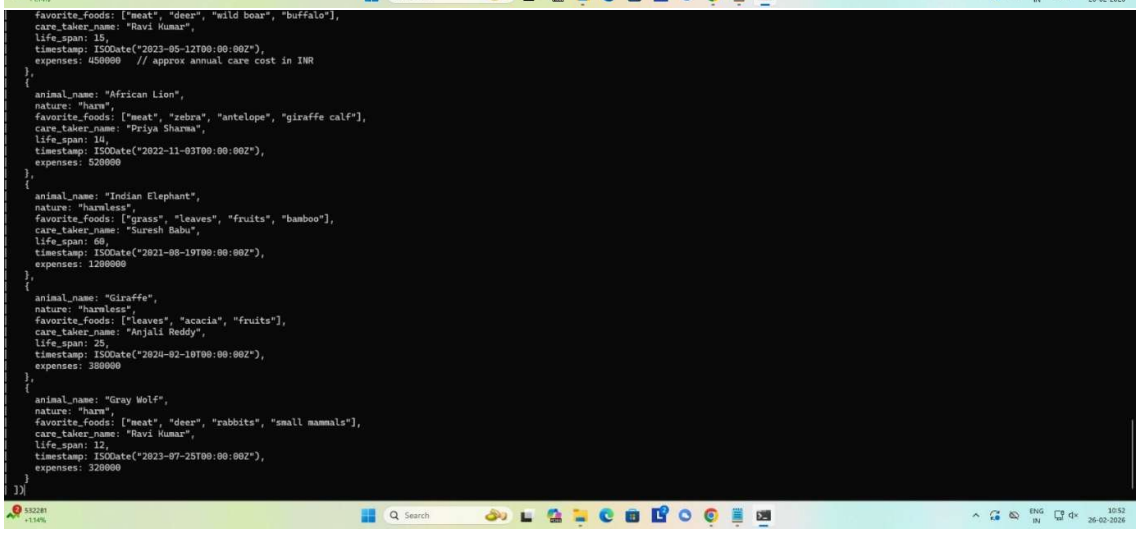
VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech **Date: 26-02-2026**
Faculty Name: Prof. S.Gopikrishnan **School: SCOPE**
Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**



```
mongosh mongodb://127.0.0.
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
|     expenses: 450000   // approx annual care cost in INR
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
animal> db.wild_animals.insertMany([
|   {
|     animal_name: "Bengal Tiger",
|     nature: "harm",
|     favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
|     care_taker_name: "Ravi Kumar",
|     life_span: 15,
|     timestamp: ISODate("2023-05-12T00:00:00Z"),
```

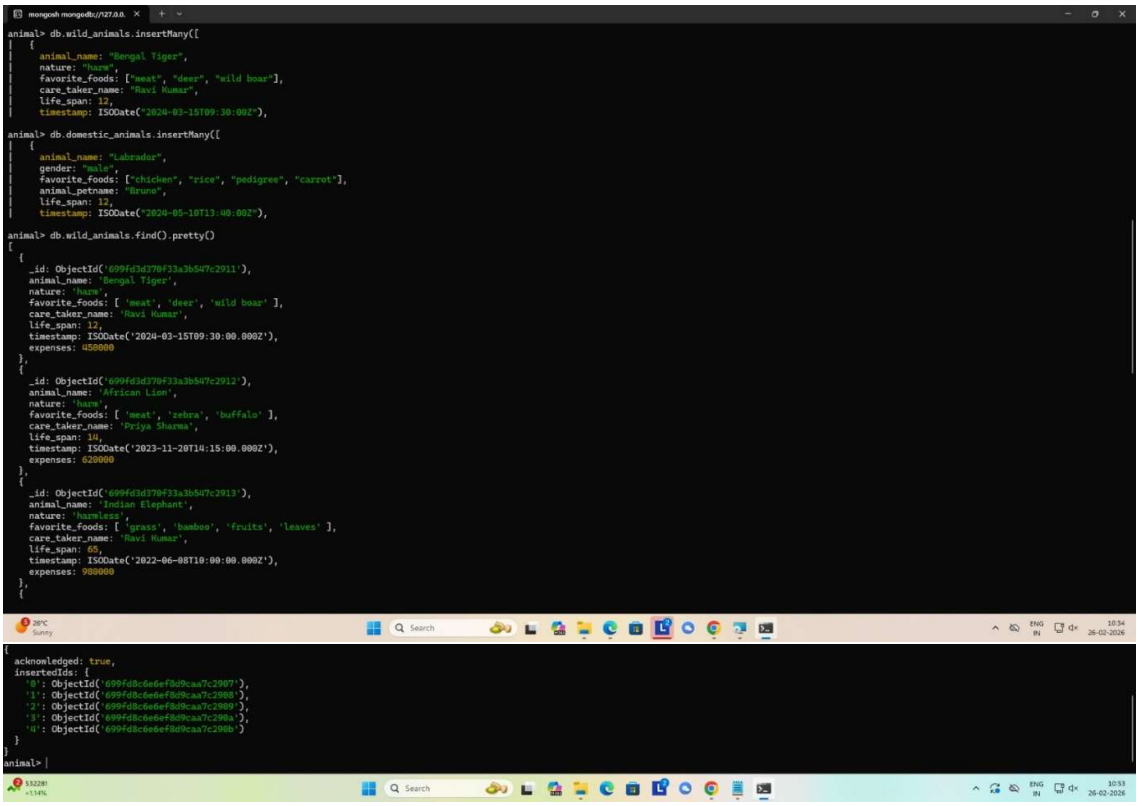


```
    favorite_foods: ["meat", "deer", "wild boar", "buffalo"],
    care_taker_name: "Ravi Kumar",
    life_span: 15,
    timestamp: ISODate("2023-05-12T00:00:00Z"),
    expenses: 450000   // approx annual care cost in INR
  },
  {
    animal_name: "African Lion",
    nature: "harm",
    favorite_foods: ["meat", "zebra", "antelope", "giraffe calf"],
    care_taker_name: "Priya Sharma",
    life_span: 14,
    timestamp: ISODate("2022-11-03T00:00:00Z"),
    expenses: 520000
  },
  {
    animal_name: "Indian Elephant",
    nature: "harmless",
    favorite_foods: ["grass", "leaves", "fruits", "bamboo"],
    care_taker_name: "Suresh Babu",
    life_span: 60,
    timestamp: ISODate("2021-08-19T00:00:00Z"),
    expenses: 1200000
  },
  {
    animal_name: "Giraffe",
    nature: "harmless",
    favorite_foods: ["leaves", "acacia", "fruits"],
    care_taker_name: "Anjali Reddy",
    life_span: 25,
    timestamp: ISODate("2024-02-10T00:00:00Z"),
    expenses: 380000
  },
  {
    animal_name: "Gray Wolf",
    nature: "harm",
    favorite_foods: ["meat", "deer", "rabbits", "small mammals"],
    care_taker_name: "Ravi Kumar",
    life_span: 12,
    timestamp: ISODate("2023-07-25T00:00:00Z"),
    expenses: 320000
  }
])
```

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech **Date: 26-02-2026**
Faculty Name: Prof. S.Gopikrishnan **School: SCOPE**
Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**



```
animal> db.wild_animals.insertMany([
  {
    animal_name: "Bengal Tiger",
    nature: "harm",
    favorite_foods: ["meat", "deer", "wild boar"],
    care_taker_name: "Ravi Kumar",
    life_span: 12,
    timestamp: ISODate("2024-03-15T09:30:00Z"),

animal> db.domestic_animals.insertMany([
  {
    animal_name: "Labrador",
    gender: "male",
    favorite_foods: ["chicken", "rice", "pedigree", "carrot"],
    animal_petname: "Bruno",
    life_span: 12,
    timestamp: ISODate("2024-05-10T13:40:00Z"),

animal> db.wild_animals.find().pretty()
[
  {
    _id: ObjectId('699fd3d370f33a3b547c2911'),
    animal_name: 'Bengal Tiger',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer', 'wild boar' ],
    care_taker_name: 'Ravi Kumar',
    life_span: 12,
    timestamp: ISODate('2024-03-15T09:30:00.000Z'),
    expenses: 450000
  },
  {
    _id: ObjectId('699fd3d370f33a3b547c2912'),
    animal_name: 'African Lion',
    nature: 'harm',
    favorite_foods: [ 'meat', 'zebra', 'buffalo' ],
    care_taker_name: 'Priya Sharma',
    life_span: 14,
    timestamp: ISODate('2023-11-20T14:15:00.000Z'),
    expenses: 620000
  },
  {
    _id: ObjectId('699fd3d370f33a3b547c2913'),
    animal_name: 'Indian Elephant',
    nature: 'harmless',
    favorite_foods: [ 'grass', 'bamboo', 'fruits', 'leaves' ],
    care_taker_name: 'Ravi Kumar',
    life_span: 65,
    timestamp: ISODate('2022-06-08T10:00:00.000Z'),
    expenses: 980000
  },
  {
```



```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699fd8c6e6ef8d9caa7c2907'),
    '1': ObjectId('699fd8c6e6ef8d9caa7c2908'),
    '2': ObjectId('699fd8c6e6ef8d9caa7c2909'),
    '3': ObjectId('699fd8c6e6ef8d9caa7c290a'),
    '4': ObjectId('699fd8c6e6ef8d9caa7c290b')
  }
}
animal>
```

5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech **Date: 26-02-2026**
Faculty Name: Prof. S.Gopikrishnan **School: SCOPE**
Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**

```
animal> db.domestic_animals.insertMany([
  {
    animal_name: "Cow",
    gender: "female",
    favorite_foods: ["grass", "hay", "grains", "vegetables"],
    animal_petname: "Lakshmi",
    life_span: 18,
    timestamp: ISODate("2022-04-15T00:00:00Z"),
    expenses: 85000
  },
  {
    animal_name: "Dog",
    gender: "male",
    favorite_foods: ["meat", "chicken", "rice", "biscuits"],
    animal_petname: "Rocky",
    life_span: 12,
    timestamp: ISODate("2023-10-08T00:00:00Z"),
    expenses: 65000
  },
  {
    animal_name: "Goat",
    gender: "female",
    favorite_foods: ["leaves", "grass", "grains"],
    animal_petname: "Meena",
    life_span: 15,
    timestamp: ISODate("2021-12-20T00:00:00Z"),
    expenses: 45000
  },
  {
    animal_name: "Horse",
    gender: "male",
    favorite_foods: ["hay", "grass", "oats", "carrots"],
    animal_petname: "Raja",
    life_span: 30,
    timestamp: ISODate("2020-06-30T00:00:00Z"),
    expenses: 150000
  },
  {
    animal_name: "Rabbit",
    gender: "female",
```

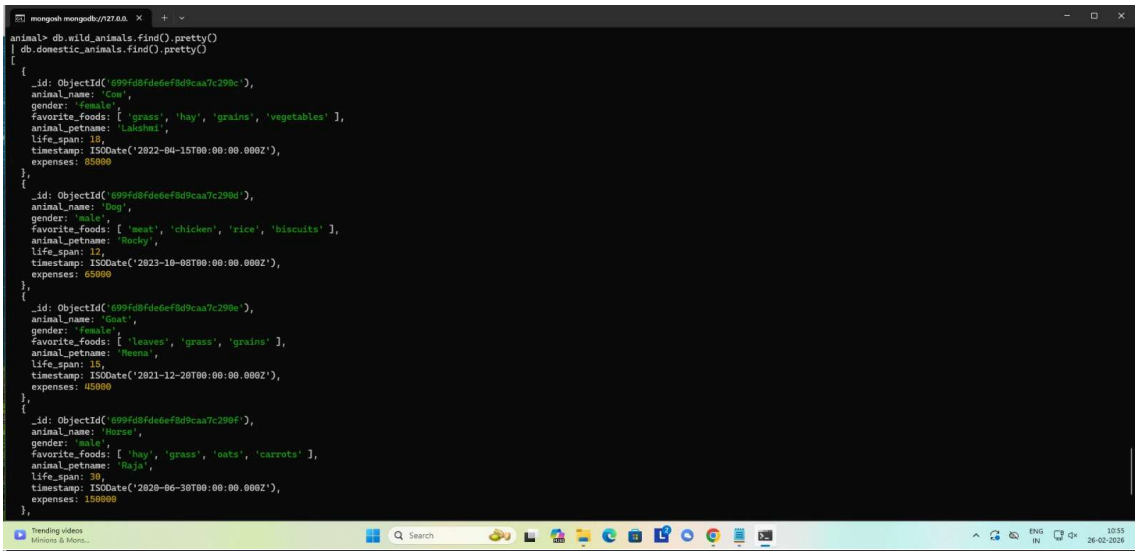
```
mongosh mongodb://127.0.0.
    life_span: 12,
    timestamp: ISODate("2023-10-08T00:00:00Z"),
    expenses: 65000
  },
  {
    animal_name: "Goat",
    gender: "female",
    favorite_foods: ["leaves", "grass", "grains"],
    animal_petname: "Meena",
    life_span: 15,
    timestamp: ISODate("2021-12-20T00:00:00Z"),
    expenses: 45000
  },
  {
    animal_name: "Horse",
    gender: "male",
    favorite_foods: ["hay", "grass", "oats", "carrots"],
    animal_petname: "Raja",
    life_span: 30,
    timestamp: ISODate("2020-06-30T00:00:00Z"),
    expenses: 150000
  },
  {
    animal_name: "Rabbit",
    gender: "female",
    favorite_foods: ["carrots", "lettuce", "hay", "pellets"],
    animal_petname: "Snowy",
    life_span: 9,
    timestamp: ISODate("2024-01-05T00:00:00Z"),
    expenses: 25000
  }
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699fd8fde6ef8d9caa7c290c'),
    '1': ObjectId('699fd8fde6ef8d9caa7c290d'),
    '2': ObjectId('699fd8fde6ef8d9caa7c290e'),
    '3': ObjectId('699fd8fde6ef8d9caa7c290f'),
    '4': ObjectId('699fd8fde6ef8d9caa7c2910')
  }
}
animal>
```

6. Write a MongoDB query to display all documents available in wild_animals and domastic_animals.

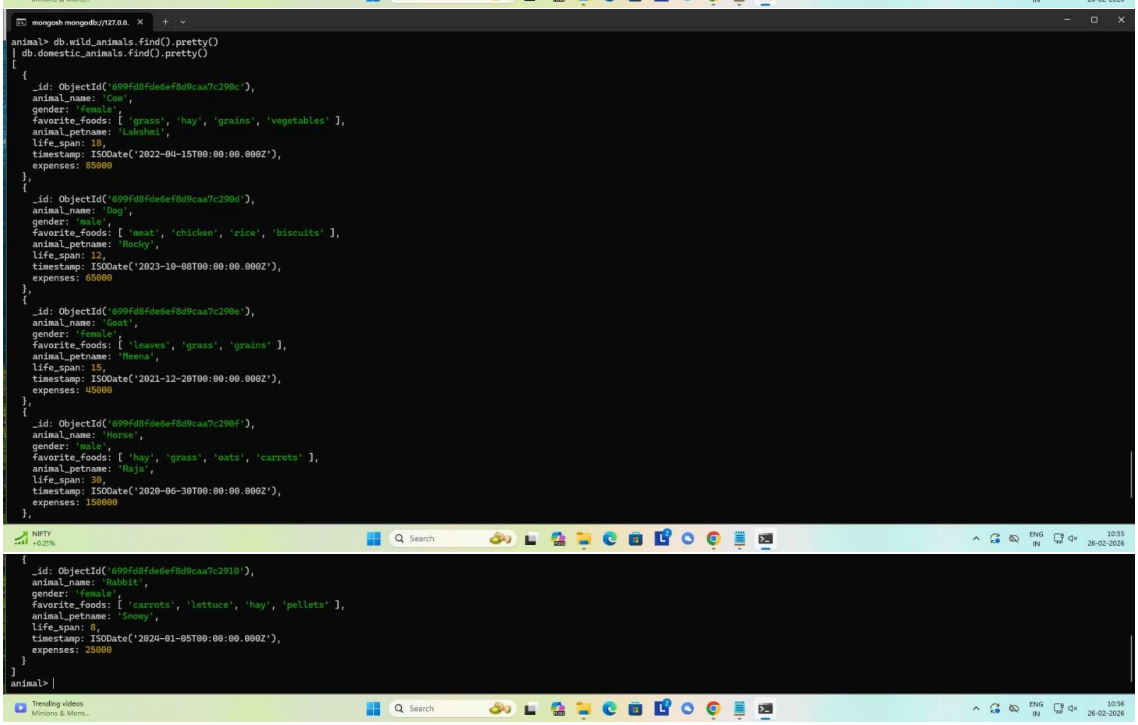
VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech **Date: 26-02-2026**
Faculty Name: Prof. S.Gopikrishnan **School: SCOPE**
Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**



```
animal> db.wild_animals.find().pretty()
| db.domestic_animals.find().pretty()
[
  {
    _id: ObjectId('699fd8fde6ef8d9caa7c290c'),
    animal_name: 'Cow',
    gender: 'female',
    favorite_foods: [ 'grass', 'hay', 'grains', 'vegetables' ],
    animal_petname: 'Lakshmi',
    life_span: 18,
    timestamp: ISODate('2022-04-15T00:00:00.000Z'),
    expenses: 85000
  },
  {
    _id: ObjectId('699fd8fde6ef8d9caa7c290d'),
    animal_name: 'Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'chicken', 'rice', 'biscuits' ],
    animal_petname: 'Rocky',
    life_span: 12,
    timestamp: ISODate('2023-10-08T00:00:00.000Z'),
    expenses: 65000
  },
  {
    _id: ObjectId('699fd8fde6ef8d9caa7c290e'),
    animal_name: 'Goat',
    gender: 'female',
    favorite_foods: [ 'leaves', 'grass', 'grains' ],
    animal_petname: 'Meena',
    life_span: 15,
    timestamp: ISODate('2021-12-20T00:00:00.000Z'),
    expenses: 45000
  },
  {
    _id: ObjectId('699fd8fde6ef8d9caa7c290f'),
    animal_name: 'Horse',
    gender: 'male',
    favorite_foods: [ 'hay', 'grass', 'oats', 'carrots' ],
    animal_petname: 'Raja',
    life_span: 30,
    timestamp: ISODate('2020-06-30T00:00:00.000Z'),
    expenses: 150000
  },
```



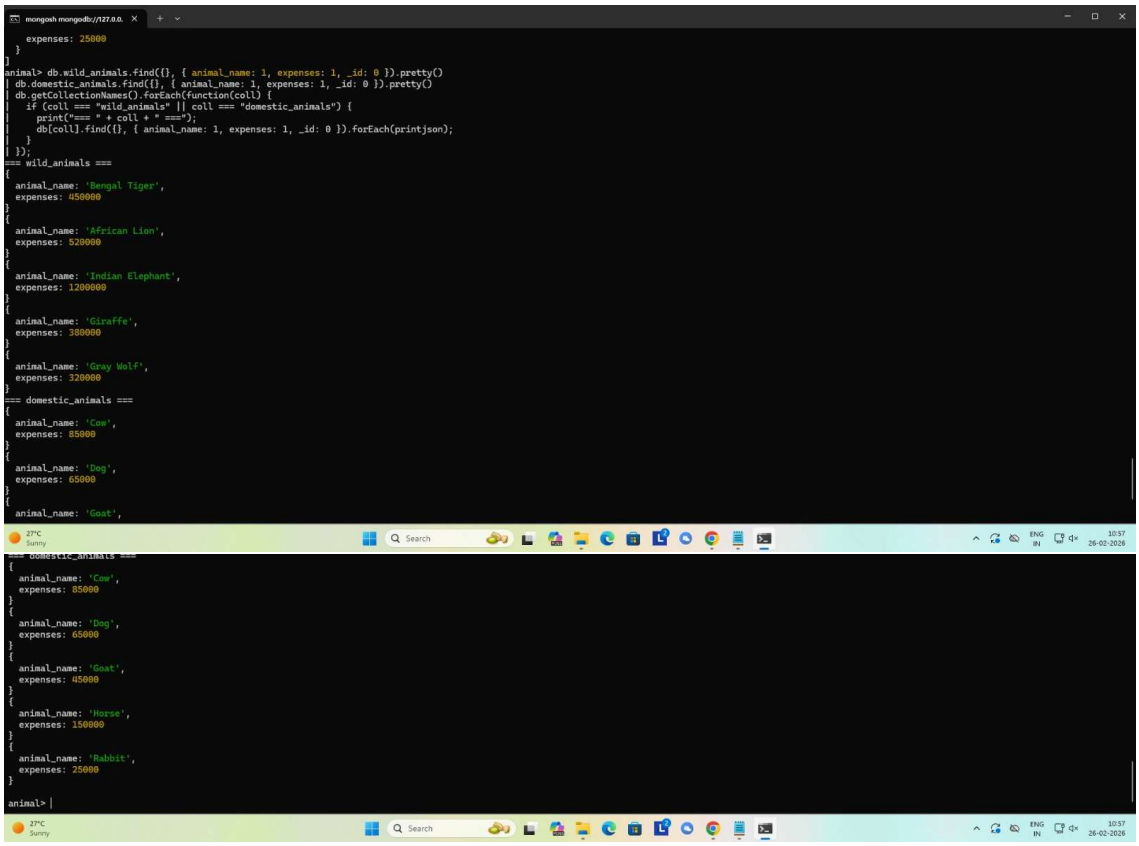
```
animal> db.wild_animals.find().pretty()
| db.domestic_animals.find().pretty()
[
  {
    _id: ObjectId('699fd8fde6ef8d9caa7c290c'),
    animal_name: 'Cow',
    gender: 'female',
    favorite_foods: [ 'grass', 'hay', 'grains', 'vegetables' ],
    animal_petname: 'Lakshmi',
    life_span: 18,
    timestamp: ISODate('2022-04-15T00:00:00.000Z'),
    expenses: 85000
  },
  {
    _id: ObjectId('699fd8fde6ef8d9caa7c290d'),
    animal_name: 'Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'chicken', 'rice', 'biscuits' ],
    animal_petname: 'Rocky',
    life_span: 12,
    timestamp: ISODate('2023-10-08T00:00:00.000Z'),
    expenses: 65000
  },
  {
    _id: ObjectId('699fd8fde6ef8d9caa7c290e'),
    animal_name: 'Goat',
    gender: 'female',
    favorite_foods: [ 'leaves', 'grass', 'grains' ],
    animal_petname: 'Meena',
    life_span: 15,
    timestamp: ISODate('2021-12-20T00:00:00.000Z'),
    expenses: 45000
  },
  {
    _id: ObjectId('699fd8fde6ef8d9caa7c290f'),
    animal_name: 'Horse',
    gender: 'male',
    favorite_foods: [ 'hay', 'grass', 'oats', 'carrots' ],
    animal_petname: 'Raja',
    life_span: 30,
    timestamp: ISODate('2020-06-30T00:00:00.000Z'),
    expenses: 150000
  },
  {
    _id: ObjectId('699fd8fde6ef8d9caa7c2910'),
    animal_name: 'Rabbit',
    gender: 'female',
    favorite_foods: [ 'carrots', 'lettuce', 'hay', 'pellets' ],
    animal_petname: 'Snowy',
    life_span: 8,
    timestamp: ISODate('2024-01-05T00:00:00.000Z'),
    expenses: 25000
  }
]
animal>
```

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech **Date: 26-02-2026**
Faculty Name: Prof. S.Gopikrishnan **School: SCOPE**
Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**



```
    expenses: 25000
  }
]
animal> db.wild_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 }).pretty()
| db.domestic_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 }).pretty()
| db.getCollectionNames().forEach(function(coll) {
|   if (coll === "wild_animals" || coll === "domestic_animals") {
|     print("=== " + coll + " ===");
|     db[coll].find({}, { animal_name: 1, expenses: 1, _id: 0 }).forEach(printjson);
|   }
| });
=== wild_animals ===
{
  animal_name: 'Bengal Tiger',
  expenses: 450000
}
{
  animal_name: 'African Lion',
  expenses: 520000
}
{
  animal_name: 'Indian Elephant',
  expenses: 1200000
}
{
  animal_name: 'Giraffe',
  expenses: 380000
}
{
  animal_name: 'Gray Wolf',
  expenses: 320000
}
=== domestic_animals ===
{
  animal_name: 'Cow',
  expenses: 85000
}
{
  animal_name: 'Dog',
  expenses: 65000
}
{
  animal_name: 'Goat',
```



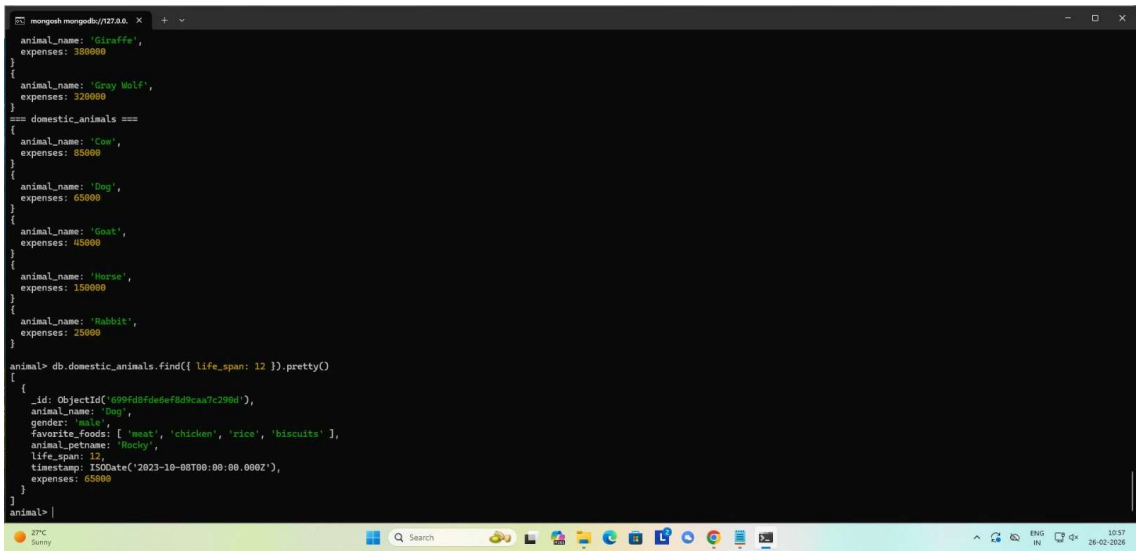
```
=== domestic_animals ===
{
  animal_name: 'Cow',
  expenses: 85000
}
{
  animal_name: 'Dog',
  expenses: 65000
}
{
  animal_name: 'Goat',
  expenses: 45000
}
{
  animal_name: 'Horse',
  expenses: 150000
}
{
  animal_name: 'Rabbit',
  expenses: 25000
}

animal>
```

8. Write a MongoDB query to display domestic_animals whose life is a particular year

VIT-AP UNIVERSITY, ANDHRA PRADESH

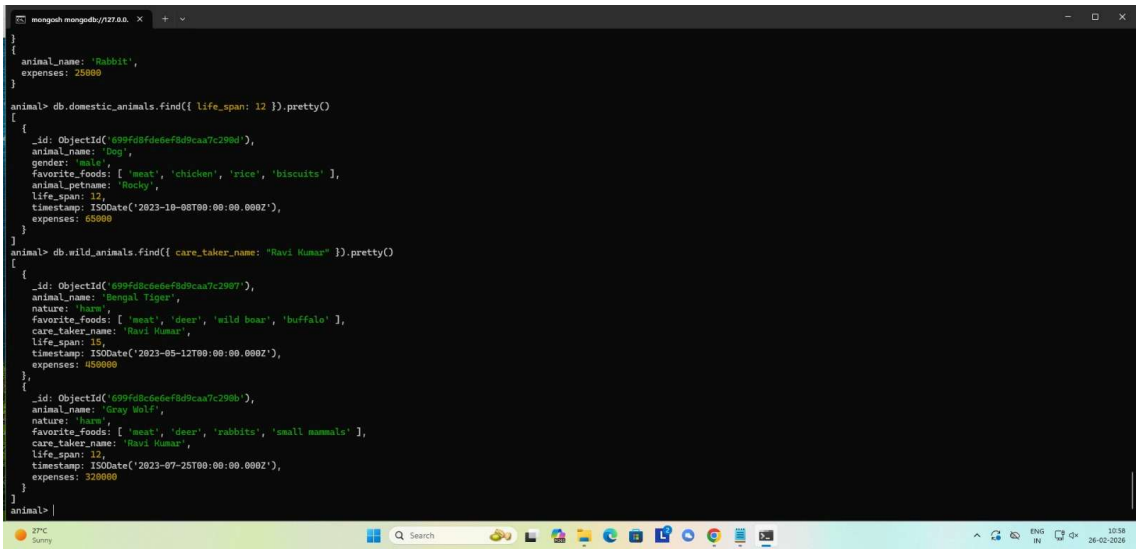
Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech **Date: 26-02-2026**
Faculty Name: Prof. S.Gopikrishnan **School: SCOPE**
Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**



```
  animal_name: 'Giraffe',
  expenses: 380000
}
{
  animal_name: 'Gray Wolf',
  expenses: 320000
}
=== domestic_animals ===
{
  animal_name: 'Cow',
  expenses: 85000
}
{
  animal_name: 'Dog',
  expenses: 65000
}
{
  animal_name: 'Goat',
  expenses: 45000
}
{
  animal_name: 'Horse',
  expenses: 150000
}
{
  animal_name: 'Rabbit',
  expenses: 25000
}

animal> db.domestic_animals.find({ life_span: 12 }).pretty()
[
  {
    _id: ObjectId('699fd8fde6ef8d9caa7c290d'),
    animal_name: 'Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'chicken', 'rice', 'biscuits' ],
    animal_petname: 'Rocky',
    life_span: 12,
    timestamp: ISODate('2023-10-08T00:00:00.000Z'),
    expenses: 65000
  }
]
animal> |
```

9. Write a MongoDB query to display wild_animals available under a particular care_taker



```
}
{
  animal_name: 'Rabbit',
  expenses: 25000
}

animal> db.domestic_animals.find({ life_span: 12 }).pretty()
[
  {
    _id: ObjectId('699fd8fde6ef8d9caa7c290d'),
    animal_name: 'Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'chicken', 'rice', 'biscuits' ],
    animal_petname: 'Rocky',
    life_span: 12,
    timestamp: ISODate('2023-10-08T00:00:00.000Z'),
    expenses: 65000
  }
]
animal> db.wild_animals.find({ care_taker_name: "Ravi Kumar" }).pretty()
[
  {
    _id: ObjectId('699fd8c6e6ef8d9caa7c2907'),
    animal_name: 'Bengal Tiger',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer', 'wild boar', 'buffalo' ],
    care_taker_name: 'Ravi Kumar',
    life_span: 15,
    timestamp: ISODate('2023-05-12T00:00:00.000Z'),
    expenses: 450000
  },
  {
    _id: ObjectId('699fd8c6e6ef8d9caa7c290b'),
    animal_name: 'Gray Wolf',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer', 'rabbits', 'small mammals' ],
    care_taker_name: 'Ravi Kumar',
    life_span: 12,
    timestamp: ISODate('2023-07-25T00:00:00.000Z'),
    expenses: 320000
  }
]
animal> |
```

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

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Student name: Chittimala Shinni Babu **Reg. no.: 23BCE20080**

---

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

```
    timestamp: ISODate('2023-07-25T00:00:00.000Z'),
    expenses: 320000
  }
]
animal> db.wild_animals.aggregate([
|   {
|     $project: {
|       animal_name: 1,
|       favorite_foods: 1,
|       expenses: 1,
|       life_span: 1,
|       type: "wild"
|     }
|   },
|   {
|     $unionWith: {
|       coll: "domestic_animals",
|       pipeline: [
|         {
|           $project: {
|             animal_name: 1,
|             favorite_foods: 1,
|             expenses: 1,
|             life_span: 1,
|             type: "domestic"
|           }
|         }
|       ]
|     }
|   },
|   { $match: { life_span: { $gt: 5 } } },
|   { $project: { animal_name: 1, favorite_foods: 1, expenses: 1, life_span: 1, type: 1, _id: 0 } }
| ]).pretty()
[
  {
    animal_name: 'Bengal Tiger',
    favorite_foods: [ 'meat', 'deer', 'wild boar', 'buffalo' ],
    life_span: 15,
    expenses: 450000,
    type: 'wild'
  },
  {
    animal_name: 'African Lion',
```

```
    animal_name: 'Gray Wolf',
    favorite_foods: [ 'meat', 'deer', 'rabbits', 'small mammals' ],
    life_span: 12,
    expenses: 320000,
    type: 'wild'
  },
  {
    animal_name: 'Cow',
    favorite_foods: [ 'grass', 'hay', 'grains', 'vegetables' ],
    life_span: 18,
    expenses: 85000,
    type: 'domestic'
  },
  {
    animal_name: 'Dog',
    favorite_foods: [ 'meat', 'chicken', 'rice', 'biscuits' ],
    life_span: 12,
    expenses: 65000,
    type: 'domestic'
  },
  {
    animal_name: 'Goat',
    favorite_foods: [ 'leaves', 'grass', 'grains' ],
    life_span: 15,
    expenses: 45000,
    type: 'domestic'
  },
  {
    animal_name: 'Horse',
    favorite_foods: [ 'hay', 'grass', 'oats', 'carrots' ],
    life_span: 30,
    expenses: 150000,
    type: 'domestic'
  },
  {
    animal_name: 'Rabbit',
    favorite_foods: [ 'carrots', 'lettuce', 'hay', 'pellets' ],
    life_span: 8,
    expenses: 25000,
    type: 'domestic'
  }
]
animal>
```